# Madhu Srinivasan, Ph.D.

Lead Scientist · Visualization and Machine Learning

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*"The purpose of computing is insight, not numbers" – Richard Hamming*

## Summary

Lead Scientist and Founding Staff of one of the largest state-of-the-art Visualization Core Laboratory at King Abdullah University of Science and Technology (KAUST). Staff Scientist at HPE Cray, pioneers of the first US Exascale systems. High Performance Computing (HPC) Visualization and Machine Learning professional with a strong expertise and over 12 years of experience in applied-research, training, and building, operating scalable workflows for visualization and large-data analysis.

## Work Experience

### HPE Cray

*Seattle, WA, USA*

Lead Scientist, Visualization and Machine Learning

*September 2019 - Present*

- Provide technical leadership within Cray's Alexandria Center of Excellence (COE). Analyze, understand, and visualize vast amounts of data proprietary to Cray's High Performance Computing systems. Act as the point of contact for the Alexandria Center for Excellence team throughout Cray.
- Leverage advanced Visualization and Machine Learning techniques to help Subject Matter Experts (SMEs) within Cray define, explore and predict systems behavior and failures. Conduct research on the data that Cray systems produce in order to increase system and application performance.
- Integrate open source analytics and AI projects, write application code, understand data science workflows, and troubleshoot and debug distributed systems. Manage multiple projects and tasks, prioritize work, and develop software projects on time.

### KAUST Visualization Core Lab

*Thuwal, Saudi Arabia*

Lead Scientist

*July 2014 - July 2019*

- Lead and direct scientific projects for large data visualization and data analysis. Manage visualization facilities, mentor and develop a team of visualization and data science experts. Collaborate with and provide support to the University's scientific communities and regional institutions.
- **Scientific Projects:** Enable scientific discovery through HPC Visualization and Data Analysis workflows in the areas of Clean Combustion Science, Marine Ecology, Earth and Climate Sciences, and Computational Fluid Dynamics (CFD).
- **Facilities:** Build and manage operations for large-scale visualization facilities such as Display Walls and CAVE Virtual Environments. The facilities represent a capital investment of around $10 million, with an annual operational budget of $1 million.
- **Project Management:** Collaborate with scientific users to gather requirements, define scope, schedule and track project milestones with my team. Follow agile project principles for providing flexible and responsive solutions to our users' unique scientific challenges.
- **Training and Scientific Outreach:** Design and administer comprehensive training sessions (5-6 annually) to train around 150-200 researchers in the field of visualization and scientific data analysis. Use visualization as a tool to explain scientific research to over 1500 visitors annually.

### KAUST Visualization Core Lab

*Thuwal, Saudi Arabia*

Visualization Scientist

*April 2009 - June 2014*

- **Scientific Projects:** Led visualization projects to build cutting-edge scientific decision-making tools for domain scientists and industry partners.
- **Software Development:** Architected solutions for large scale data visualization and interaction with distributed graphics on ultra-high resolution tiled-displays and virtual reality environments.
- **HPC Visualization:** Led efforts to design and implement *in situ* and streaming High Performance Computing (HPC) Visualization from supercomputing facilities at KAUST.
- **Training:** Trained and Mentored over 200 researchers annually, in using various tools and frameworks for data visualization.

### Laika Studios, Inc.

*Portland, OR, USA*

Technical Director

*August 2008 - December 2008*

- Worked with the Front-end Pipeline team to develop tools used by 3D artists and animators in the studio.
- Responsibilities included developing front-end GUI for 3D/2D Digital Asset Management, Maya/Houdini Integration.
- Coordinated daily with production department heads to collect requirements and feature-requests, and provide solutions around their established workflows, using agile principles.
- Negotiated project scope and feature additions, debugged production-related technical problems, supported 3D artists with various aspects of the production pipeline.

### Google, Inc.

*Mountain View, CA, USA*

Engineering Research Intern

*January 2007 - March 2007*

- Researched various statistical models for enhancing resolution of photographic images.
- Implemented "Statistical Image Super-resolution" using OpenCV and MapReduce technologies.

## Skills

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|---|---|
| **Programming Languages** | Python, Rust, Javascript, Elixir, C/C++ |
| **Data Visualization and 3D Graphics** | D3js, Highcharts/AMCharts, Paraview/VTK, Avizo3D, Bokeh, HoloViews, Blender 3D, Unity3D Engine |
| **Web and Databases** | Node.js, Phoenix Framework, React.js, HTML5/WebGL, Webpack, OmniSci Core |
| **Distributed Computing** | MPI, DisplayCluster, Erlang OTP, Paraview/Catalyst, Dask |
| **Data Science** | Anaconda, Pandas, SciKit-Learn, Jupyter Lab |

## Articles and Publications

- In Situ Workflow Specification. M. Parashar, P. Bremer, K. Heitmann, M. Larsen, J. Patchett, T. Peterka, **M. Srinivasan**, N. Röber, S. Frey, and B. Raffin, In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 13, 2019.
- Software Complexity, Heterogeneity, and User-facing Issues. E. Bethel, J. Patchett, P. Messmer, M. Parashar, A. Bauer, **M. Srinivasan**, K. Ono, N. Röber, N. Gauger, J. Bennett, et al., In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 34, 2019.
- RedSeaAtlas: A Visual Analytics Tool for Spatiotemporal Multivariate Red Sea Data. S. Afzal, S. Ghani, G. Tissington, S. Langodan, T. Jamil, **M. Srinivasan**, and I. Hoteit. Submitted to the Workshop on Visualization in Environmental Sciences 2019 (EnvirVis 2019)
- Sonification of Animal Tracks as an Alternative Representation of Multi-Dimensional Data: A Northern Elephant Seal Example. Duarte CM, Riker P, **Srinivasan M**, Robinson PW, Gallo-Reynoso JP, et al. (2018).Frontiers in Marine Science.
- Simulation and Visualization of the Cyclonic Storm Chapala over the Arabian Sea: A Case Study. T. Theußl, H. P. Dasari, I. Hoteit and **M. Srinivasan**, 4th Saudi International Conference on Information Technology (IEEE Big Data Analysis) (KACSTIT), Riyadh 2016.
- Three-dimensional immersive virtual reality for studying cellular compartments in 3D models from EM preparations of neural tissues. Corrado Cali, Daniya Boges, Jumana Baghabra, Glendon Holst, Anna Kreshuk, Fred Hamprecht, **Madhusudhanan Srinivasan**, Heikki Lehvaslaiho, Pierre Magistretti. Journal of Comparative Neurology, 2015.
- PathlinesExplorer - Image-based exploration of large-scale pathline fields. O. H. Nagoor, M. Hadwiger and **M. Srinivasan**. 2015 IEEE Scientific Visualization Conference (SciVis), Chicago, IL, 2015, pp. 159-160 (Video Preview)
- Query-driven Parallel Exploration of Large Datasets. Atanas Atanasov, **Madhusudhanan Srinivasan**, Tobias Weinzierl, In proceedings of IEEE Symposium on Large-Scale Data Analysis and Visualization, Seattle, WA (2012).
- Semi-automatic end-user tools for construction of virtual avatar behaviors. **Madhusudhanan Srinivasan** and Ronald A. Metoyer, In Proceedings of the 16th International Conference on 3D Web Technology (Web3D 2011). ACM, New York, NY, USA, 121-128. (Video Preview)
- Acquisition of stereo panoramas for display in VR environments. Richard A. Ainsworth, Daniel J. Sandin, Jurgen P. Schulze, Andrew Prudhomme, Thomas A. DeFanti and **Madhusudhanan Srinivasan**, Proc. SPIE 7864, 786416 (2011).
- The Future of the CAVE. DeFanti, T.A., Acevedo, D., Ainsworth, R.A., Brown, M.D., Cutchin, S., Dawe, G., Doerr, K., Johnson, A.E., Knox, C. Kooima, R., Kuester, F. Leigh, J., Long, L., Otto, P., Petrovic, V., Ponto, K., Prudhomme, A., Rao, R., Renambot, L., Sandin, D.J., Schulze, J.P., Smarr, L., **Srinivasan, M.**, Weber, P., Wickham, G. Central European Journal of Engineering 2011, Pages 16-37.
- Towards a Digital Makkah - Using Immersive 3D Environments to Train and Prepare Pilgrims. Jens Schneider, Dina Garatly, **Madhusudhanan Srinivasan**, Stephen Guy, Sean Curtis, Steven Cutchin, Dinesh Manocha, Ming Lin, Alyn Rockwood. International Conference on Digital Media and its Applications in Cultural Heritage, March 2011.
- Multisource Reverse-Time Migration and Full-Waveform Inversion on a GPGPU. C. Boonyasiriwat, G. Zhan, M. Hadwiger, **M.Srinivasan**, and G. T. Schuster. 72nd EAGE Conference and Exhibition incorporating SPE EUROPEC 2010
- Controllable real-time locomotion using mobility maps. **Madhusudhanan Srinivasan**, Ronald A. Metoyer, and Eric N. Mortensen. In Proceedings of Graphics Interface 2005 (GI '05). Canadian Human-Computer Communications Society, School of Computer Science, University of Waterloo, Waterloo, Ontario, Canada, 51-59.
- A Data-Driven Model of Pedestrian Movement. 3rd International Conference on Pedestrian and Evacuation Dynamics. Casburn, L., **Srinivasan M.**, Metoyer, R., and Quinn, M. Vienna, Austria. pp. 189-200
- A Tangible Interface for High-Level Direction of Multiple Animated Characters. Metoyer, R. A.; Xu, L. and **Srinivasan, M.**. In Proceedings of Graphics Interface 2003, pp. 167-176.
- A Responsiveness Metric for Controllable Characters. **M. Srinivasan** and R. Metoyer, School of Electrical Engineering and Computer Science, Oregon State University, Technical Report #CS05-50-01.

## Awards, Distinctions & Invited Talks

**Dagstuhl Seminar: *In Situ* Visualization for Computational Science** — *Dagstuhl, Germany*

Invited Contributor & Participant — *July 2018*

**Leading@KAUST Program by Harvard Business Publishing** — *KAUST, Saudi Arabia*

Nominated Participant — *March 2015*

**Workshop on Visualization Infrastructure & Systems Technology** — *New Orleans, U.S.A*

Invited Speaker: KAUST Visualization Laboratory - Sustaining and Operating at Scale — *November 2014*

**Recognition for Distinguished Service** — *KAUST, Saudi Arabia*

Five-year Service Award presented by the President of the University — *May 2014*

**Microsoft Research eScience Workshop** — *Stockholm, Sweden*

Invited Panel Speaker: *Is NUI "Natural" for Scientists?* — *December 2011*

## Program Committees

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| 2019 | **Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization | *Porto, Portugal* |
| 2018 | **Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization | *Brno, Czech* |
| 2017 | **Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization | *Barcelona, Spain* |
| 2016 | **Program Committee**, IEEE Symposium on Large Data Analysis and Visualization | *Baltimore, MD* |
| 2014 | **Program Committee**, IEEE Symposium on Large Data Analysis and Visualization | *Paris, France* |
| 2013 | **Program Committee**, IEEE Symposium on Large Data Analysis and Visualization | *Atlanta, GA* |
| 2017 | **Organizing Committee**, High Performance Computing - Saudi Arabia | *Saudi Arabia* |
| 2015 | **Planning Committee**, Supercomputing Conference | *Austin, TX* |
| 2014 | **Planning Committee**, Supercomputing Conference | *New Orleans, LA* |
| 2015 | **Co-Organizer**, Workshop on Visualization Infrastructure and Systems Technology | *Austin, TX* |

## Mentoring and Advising

### Thesis Advising

**Ph.D. Thesis: Mr. T. Muralidharan** — *MGR Research Institute, India*

Hybrid Extensive Machine Learning Algorithm For Domain Specific Features To Detect And Classify Web Spam — *March 2019*

**Master's Thesis: Ms. Omniah Nagoor** — *KAUST, Saudi Arabia*

Image-based Exploration of Large-Scale Pathline Fields — *May 2014*

**Master's Thesis: Ms. Roba Binyahib** — *KAUST, Saudi Arabia*

Image-based Exploration of Iso-surfaces for Large Multi-Variable Datasets using Parameter Space — *May 2013*

**Research Mentor** — *KAUST, Saudi Arabia*

Saudi Research Science Institute Program — *July 2016-2018*

**Training: HPC Visualization Tools** — *Jeddah, Saudi Arabia*

High Performance Computing – Saudi Arabia — *March 2017*

## Education

**Ph.D. Computer Science** — *July 2005 - March 2009*

Oregon State University, College of Engineering — *Oregon, USA*

GPA 3.91/4.00

**M.S. Computer Science** — *September 2001 - June 2005*

Oregon State University, College of Engineering — *Oregon, USA*

GPA 3.90/4.00

**B.Engg. Computer Science** — *July 1997 - May 2001*

Maharaja Sayajirao University, Faculty of Engineering and Technology — *Baroda, India*

GPA 3.99/4.00. First Class with Distinction